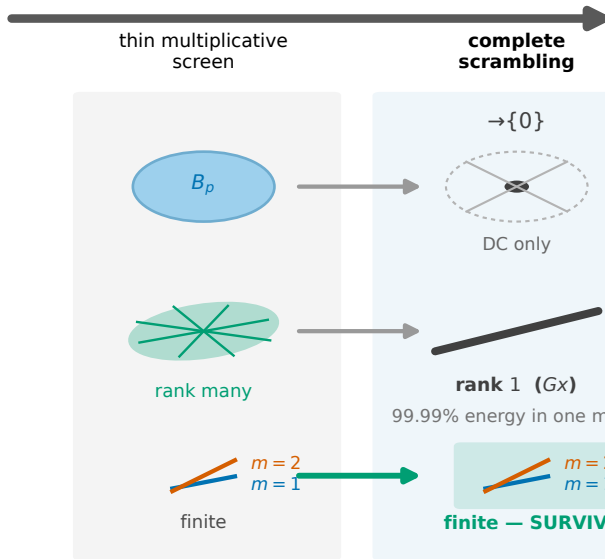


OPTICAL DISORDER



complete-scrambling law

$$\text{Cov}(b) = Q(x) (O \circ O) + R$$

$$Q(x) = x^\top G x, \quad G > 0$$

two directional classes:

$$x^\top G \delta \neq 0 \Rightarrow m = 1$$

first-order observable

$$x^\top G \delta = 0 \Rightarrow m = 2$$

curvature-rescued ($\delta^\top G \delta > 0$)

⇒ blind set empty



amplitude anchor required
 (else quotient info $J \rightarrow 0$)

Strong disorder erases **coordinates** (support $\rightarrow \{0\}$, rank $\rightarrow 1$) but not **local change observability**: the quotient jet order stays finite.